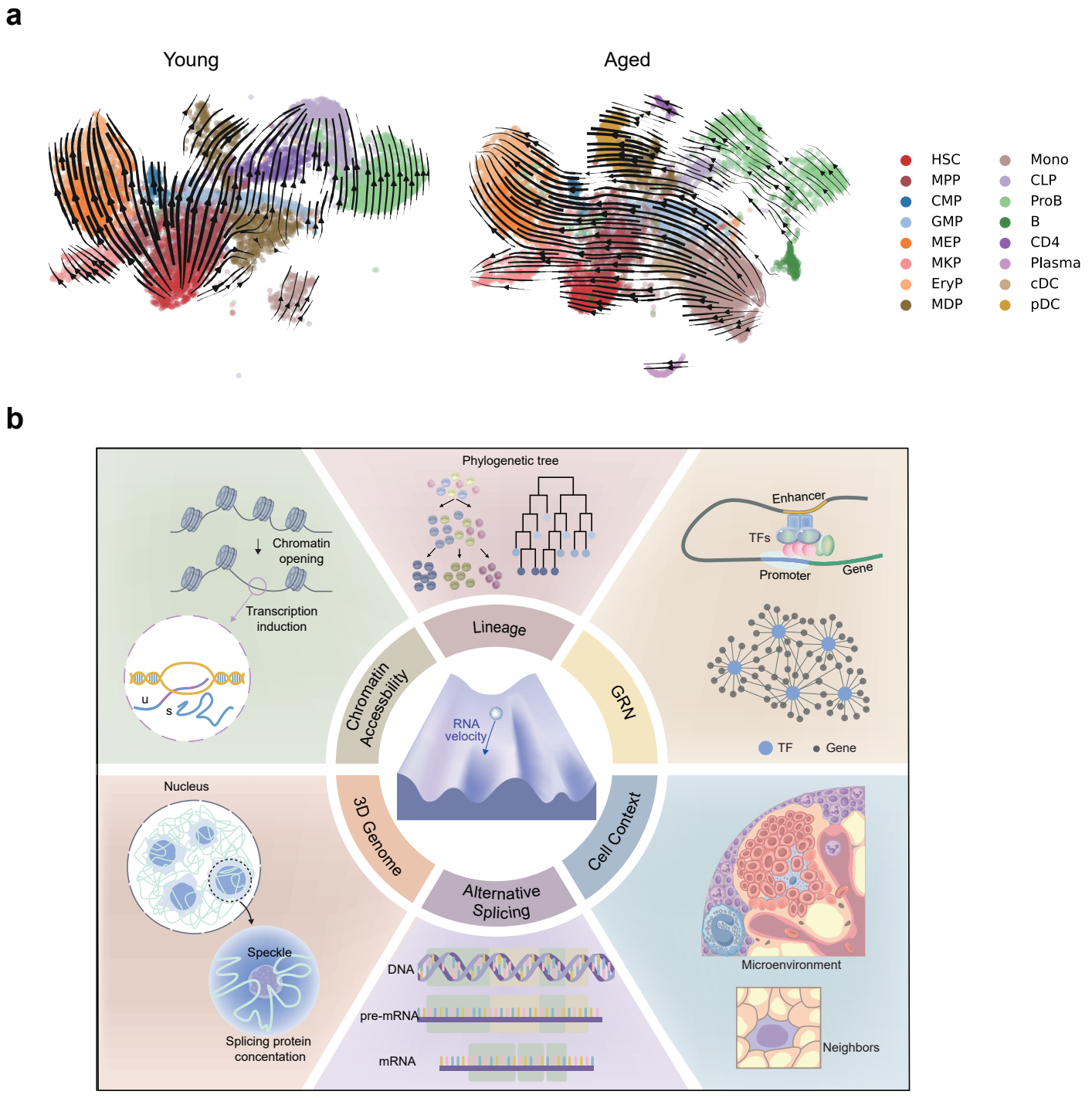


Fig. 7 | Development potential and future directions of RNA velocity. a, RNA velocity stream plots on the ReDeeM datasets of young and aged human hematopoiesis, respectively. Velocities were inferred using LatentVelo. cDC, conventional dendritic cell; pDC, plasmacytoid dendritic cell. **b,** Conceptual perspectives that have already been integrated with RNA velocity inference, as well as those that may be incorporated in the future. The central panel illustrates the classical Waddington landscape, in which a cell, represented as a pebble, starts at the top of a hill and follows one of several possible developmental paths. In this analogy, RNA velocity can be viewed as the sliding velocity of the pebble, indicating its future direction. The upper panel highlights modalities already incorporated by existing RNA velocity methods, including chromatin accessibility, lineage information, and gene regulatory networks (GRNs). The lower panel illustrates additional perspectives that may be incorporated in future methods, including 3D genome organization, alternative splicing, and cell context.